// MISSION BRIEFING

ASTROPIXEL MISSION DESIGNER

Build a spacecraft. Fly the whole mission.

See how every part depends on every other.

TEAM SHAY RAFE3 | SPACE MISSION DESIGN GAME | RAMALLAH

Omar · Ahmad · Ameen · Mahdi



NASA SPACE APPS CHALLENGE 2026

Millions love space. Few see how a mission works

149M

views of NASA's Artemis II coverage, 2026

222M

people follow NASA on social media

114,000+

people joined NASA Space Apps in 2025

BUT TODAY, ALL THEY CAN DO IS...



WATCH

Livestreams show the launch, not the hundreds of choices behind it



READ

Courses teach one topic at a time, mostly at university



PLAY

Space games fly rockets, with no budget, power, data or team

Nothing lets you run a real mission yourself, from the first idea to the final result.

Sources: NASA (Artemis II streaming record, 2026; NASA social media, 2023); NASA Space Apps 2025 season

IMPACT

In Palestine, the gap is even wider



No aerospace engineering degree

that we could find at a Palestinian university



A trip to NASA for a lucky few

the best-known path takes a small group of teens to NASA Goddard



Hands-on space learning is rare

mostly small volunteer astronomy clubs, not mission design

The knowledge exists.

The access doesn't.

CREATIVITY RELEVANCE

AstroPixel: a whole mission, in any browser



The hangar: snap real parts on. Mission Control lamps go green only with NASA's margins.

1 BUILD

Pick a world and build from 29 real parts. Every resource is tracked live

2 FLY

Launch, cruise through real solar storms, land in real gravity

3 FAIL SAFELY

The report traces the cause chain: what broke, and why

4 FIX AND FLY AGAIN

Alone, or as 4 roles who must all sign off, like a NASA team

WHAT PLAYERS DISCOVER +1 instrument ▸ more power ▸ more mass ▸ more fuel ▸ bigger rocket ▸ more money

RELEVANCE VALIDITY

Four real rules decide every mission



ROCKET EQUATION

$$\Delta v = I_{sp} \cdot g_0 \cdot \ln(m_0/m_1)$$

Every extra kg needs more fuel

NASA GMAT · engine Isp 300 s



SUNLIGHT ∝ 1/d²

Saturn: 14.8 W/m² ≈ 1%

Far out: nuclear MMRTG, 110 W

NSSDCA Fact Sheets · NASA MMRTG



RADIO ∝ 1/d²

Mars 4–24 min · Saturn ~80

Big dish; landers land themselves

NSSDCA · MRO 3 m dish



NASA MARGINS

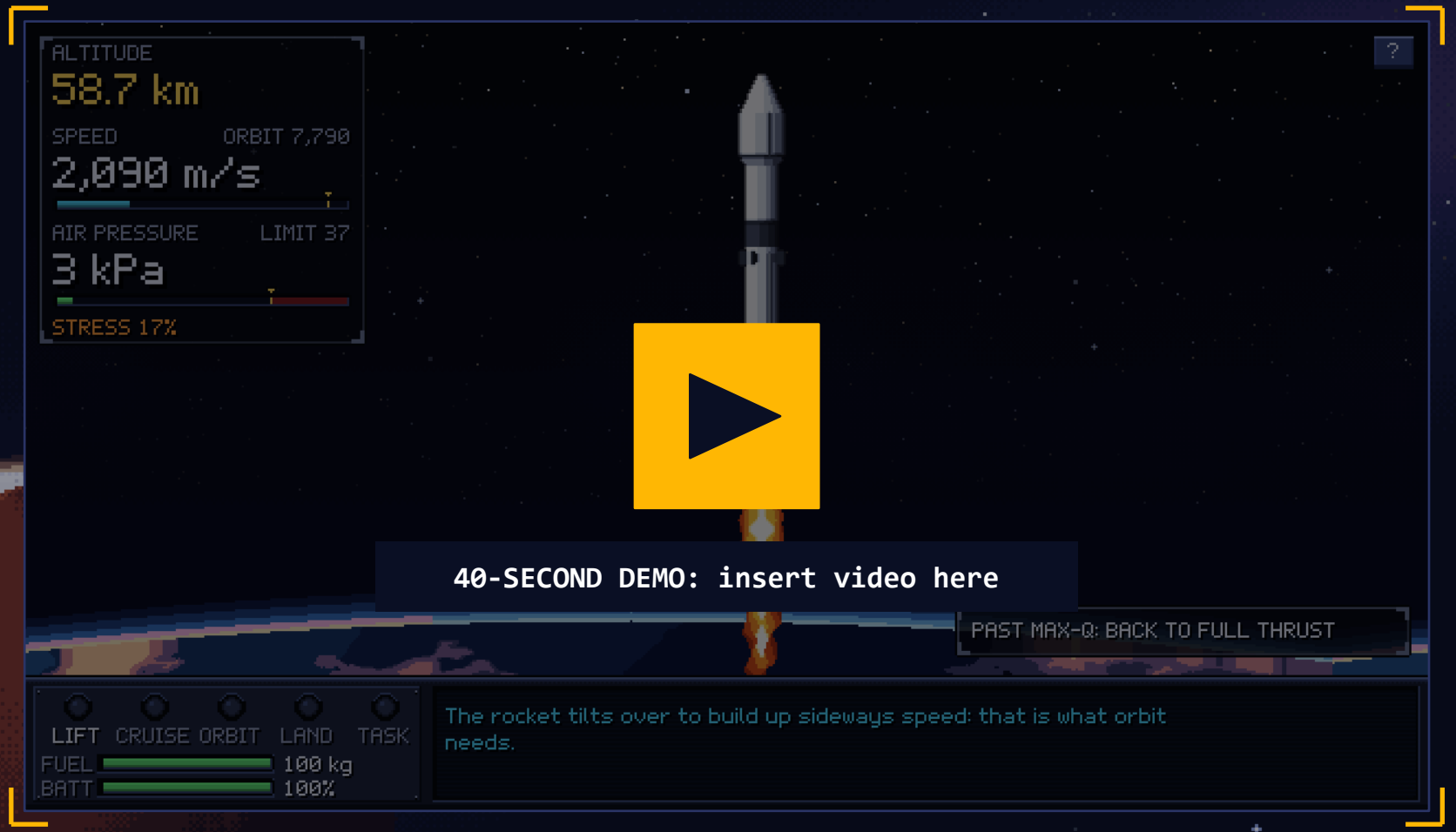
keep 35% spare

Green only with JPL's margin

JPL CML 4 · 3 dB link · 25% cost

REAL EVENTS NASA DONKI storms (GST, 2024), e.g. 10 May 2024, Kp 9. Every number has a source.

See it fly



SCAN TO PLAY

free · no install · no login

- 8 worlds × 3 tasks
- 1–4 players, one role each

omarshobaki-png.github.io/Shay-Rafe3-Nasa-Space-App-Challenge/play/

Lose your first spacecraft here, for free



Students and space fans

Learn mass, power, data and comms by using them, from zero knowledge

Teachers and schools

One free link for a class: teams of 4 roles, a report that explains every result

Palestinian learners

Any browser on a school PC: no install, no account, no travel

NEXT STEPS

1. Pilot with one school class in Ramallah
2. Fix what first-time players trip on
3. Add real NASA photos (LRO, HiRISE, Cassini)

HONEST LIMITS

Robotic missions only · flight paths from real missions, not full GMAT runs · English only for now

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AI disclosure: the idea and game design are ours; AI helped write code and draft this deck. No AI images, video or audio: all art comes from our own scripts.